

12. QUANTUM ESTIMATION THEORY

Estimation theory is the study of how to make efficient estimates of unknown parameters from a given set of data. Imagine that we are interested in some parameter, θ , that we cannot directly measure. However, through some given rule, θ influences a random variable, X , that we can observe. By sampling from X we try to estimate θ . For example, θ could be the probability of observing Heads when flipping a coin and X could be the random variable associated with each coin flip. By flipping the coin many times, we could make increasingly precise estimates of θ . In what follows, we introduce the necessary background in estimation theory and then discuss how quantum resources can improve our ability to make more precise estimations.

12.1. Estimators

Consider that we are given a set of data $x = \{x_1, \dots, x_N\}$ which consists of independent samples, x_i , from identically distributed random variables, X_i . The parameter θ influences $X = (X_1, X_2, \dots)$ according to a given model and therefore the probability distribution of X can be written $p_X(x|\theta)$. We must convert our data into a guess for the value of θ ⁷⁵. This inference rule is called an *estimator* and we write it as $\hat{\theta}(x)$. For convenience, we sometimes omit the dependence on x and just write $\hat{\theta}(x) = \hat{\theta}$.

How should we choose our estimator? To answer this, we need a notion of what makes for a good estimator. Naturally, a good estimator should output estimates that are near the true parameter value. It is common to quantify this via the mean-square error,

$$(\Delta\hat{\theta})^2 \equiv E[(\hat{\theta} - \theta)^2] = \int dx p(x|\theta) (\hat{\theta} - \theta)^2. \quad (12.1)$$

A good estimator has a small mean-square error, but this quantity is θ -dependent so the estimator might be good around some values of θ and bad for other values of θ . Typically, we assume that we have a rough idea of what θ should be.

A convenient property to have in an estimator is that it is *unbiased*, namely that it on average returns the true value of θ ,

$$E[\hat{\theta}] = \int dx p(x|\theta) \hat{\theta}(x) = \theta. \quad (12.2)$$

Notice that for an unbiased estimator, the mean-squared error is the same as the estimator's variance because

$$E[(\hat{\theta} - \theta)^2] = E[\hat{\theta}^2] + \theta^2 - 2\theta \underbrace{E[\hat{\theta}]}_{=\theta} = E[\hat{\theta}^2] - \theta^2 = \text{Var}[\hat{\theta}]. \quad (12.3)$$

However, unbiased estimators are not always available for a given estimation problem, and even if they are it is not always immediate how to find them. Let us illustrate the latter with a simple example.

12.1.1. Example: Normal distribution

Let $X_1, \dots, X_N$ be independent and normally distributed random variables with expectation value μ and variance σ^2 . Consider now that we want to estimate the variance. An intuitive guess for a good estimator is the average mean-square deviation between each random variable and its sample average,

$$\hat{\theta} = \frac{1}{N} \sum_{i=1}^N (X_i - \bar{X})^2, \quad (12.4)$$

where $\bar{X} = \frac{1}{N} \sum_i X_i$ is the sample average. To check whether the estimator is unbiased, we compute the expectation value

$$E[\hat{\theta}] \stackrel{(12.4)}{=} \frac{1}{N} \sum_{i=1}^N E[(X_i - \bar{X})^2] = \frac{1}{N} \sum_{i=1}^N E\left[\left(\frac{1}{N} \sum_k (X_i - X_k)\right)^2\right] = \frac{1}{N^3} \sum_{i=1}^N E\left[\sum_{j,k} (X_i - X_k)(X_i - X_j)\right]. \quad (12.5)$$

⁷⁵ We use θ interchangeably for the parameter to be estimated and the true value of that parameter.

We can now expand the right-most expectation value. Note that since the random variables are independent and identically distributed we have $E[X_i X_j] = E[X_i]E[X_j]$ and $E[X_i] = E[X_1] \forall i$. Using these relations we get

$$\begin{aligned} E \left[\sum_{j,k} (X_i - X_k)(X_i - X_j) \right] &= \sum_{j,k} E [X_i^2 + X_j X_k - X_i X_k - X_i X_j] = N^2 E [X_i^2] + \left(\sum_j E [X_j^2] + \sum_{j \neq k} E [X_j X_k] \right) \\ &\quad - \left(\sum_j E [X_i^2] + N \sum_{k \neq i} E [X_i X_k] \right) - \left(\sum_k E [X_i^2] + N \sum_{j \neq i} E [X_i X_j] \right) \\ &= N^2 E [X_1^2] + (N E [X_1^2] + N(N-1) E [X_1]^2) - (N E [X_1^2] + N(N-1) E [X_1]^2) \\ &\quad - (N E [X_1^2] + N(N-1) E [X_1]^2) = N(N-1) (E [X_1^2] - E [X_1]^2) = N(N-1) \sigma^2, \end{aligned} \quad (12.6)$$

where we have written $\sigma^2 = \text{Var}[X_1]$. Inserted into Eq (12.5) we obtain

$$E[\hat{\theta}] = \frac{N-1}{N} \sigma^2. \quad (12.7)$$

For the estimator to be unbiased we would have needed $E[\hat{\theta}] = \sigma^2$. Hence, our estimator is biased by a factor $\frac{N-1}{N}$. However, this factor is only significant for small data sets, as it tends to one with large N . In contrast, if we wanted to make an unbiased estimator for every N , we would have had to make a not-so-intuitive modification to Eq (12.4), namely replacing it with

$$\hat{\theta} = \frac{1}{N-1} \sum_{i=1}^N (X_i - \bar{X}). \quad (12.8)$$

Repeating the above calculation with this estimator, one finds that $E[\hat{\theta}] = \sigma^2$. This estimator is called the *unbiased sample variance* and the correction factor of $\frac{N}{N-1}$ is called Bessel's correction.

12.1.2. Maximum likelihood estimator

A popular systematic way of selecting estimators is the so-called *maximum likelihood estimator*. It tells us to first draw x and then guess on the θ that assigns the highest probability to our sample.

G26: Maximum likelihood estimator

Let the random variable X depend on the parameter θ and denote its conditional probability distribution $p_X(x|\theta)$. The maximum likelihood estimator is

$$\hat{\theta}(x) = \underset{\theta}{\operatorname{argmax}} p_X(x|\theta). \quad (12.9)$$

Actually, it is often convenient to work not directly with the likelihood function, namely $p_X(x|\theta)$, but instead with its logarithm. The log-likelihood function is therefore $\ln p_X(x|\theta)$ ⁷⁶. Note that since the logarithm is strictly increasing its argmax will remain the same as in Eq (12.9). The reason that one may prefer the log-likelihood function is that it is an additive quantity thanks to $\ln(xy) = \ln(x) + \ln(y)$. Let us explore the maximum log-likelihood estimator in a couple of concrete cases.

Example: Bernoulli distribution.— Consider a binary random variable, X , with distribution $p_X(0) = 1 - \theta$ and $p_X(1) = \theta$. Given data $x = \{x_1, \dots, x_N\}$, the likelihood function is

$$p_X(x|\theta) = \theta^{\#ones} (1 - \theta)^{\#zeros} = \theta^{\sum_i x_i} (1 - \theta)^{N - \sum_i x_i} = \theta^{N\bar{x}} (1 - \theta)^{N(1-\bar{x})}, \quad (12.10)$$

where we have defined the sample average $\bar{x} = \frac{1}{N} \sum_i x_i$. The log-likelihood function then becomes

$$\ln p_X(x|\theta) = N\bar{x} \ln(\theta) + N(1 - \bar{x}) \ln(1 - \theta). \quad (12.11)$$

⁷⁶ Here we use the natural logarithm.

We want to select the θ that maximises the log-likelihood. For that we find the roots of the derivative

$$0 = \frac{\partial \ln p_X(x|\theta)}{\partial \theta} = N \left(\frac{\bar{x}}{\theta} - \frac{1 - \bar{x}}{1 - \theta} \right) \Rightarrow \hat{\theta}(x) = \bar{x}. \quad (12.12)$$

Thus, our maximum likelihood estimator is just the relative frequency of observing the event “1”⁷⁷. We can also confirm that this estimator is unbiased because

$$E[\hat{\theta}] = \frac{1}{N} \sum_{i=1}^N E[X_i] = \frac{1}{N} \sum_{i=1}^N \theta = \theta. \quad (12.13)$$

Example: Normal distribution.— Let X_i be independent and normally distributed random variables with mean θ and variance σ^2 . The likelihood function is

$$p_X(x|\theta, \sigma^2) = \prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \theta)^2}{2\sigma^2}\right). \quad (12.14)$$

The log-likelihood function becomes

$$\ln p_X(x|\theta, \sigma^2) = -\sum_i \frac{(x_i - \theta)^2}{2\sigma^2} - \frac{N}{2} \ln(2\pi\sigma^2). \quad (12.15)$$

Say that our goal is to estimate θ via the maximum likelihood estimator. Then the most likely value of θ is that which maximises the log-likelihood function. Hence we solve

$$0 = \frac{\partial \ln p_X(x|\theta, \sigma^2)}{\partial \theta} = \frac{N(\bar{x} - \theta)}{\sigma^2} \Rightarrow \hat{\theta}(x) = \bar{x}. \quad (12.16)$$

where we have again set $\bar{x} = \frac{1}{N} \sum_i x_i$. Thus, our maximum likelihood estimator is just the sample average.

Consider that we instead want to estimate the variance, σ^2 . For that, we again follow the maximum likelihood approach to get

$$0 = \frac{\partial \ln p_X(x|\theta, \sigma^2)}{\partial \sigma^2} = \frac{N}{\sigma^4} \left(\sigma^2 - \frac{1}{N} \sum_i (x_i - \theta)^2 \right) \Rightarrow \hat{\sigma}^2 = \frac{1}{N} \sum_i (x_i - \theta)^2. \quad (12.17)$$

If we were to substitute θ for our maximum likelihood estimate, namely $\hat{\theta} = \bar{x}$, we recover the average sample variance in Eq (12.4), but without Bessel’s correction. Consequently, the maximum likelihood estimator is not always unbiased for finite N .

12.2. Cramér-Rao bound and Fisher information

How good can we ever hope for an estimator to be? The *Cramér-Rao bound* provides a lower bound on the mean-square error for any unbiased estimator. Thus, it sets a limit for how well we could make our estimations.

R27: Cramér-Rao bound

Let $\hat{\theta}(x)$ be an unbiased estimator for θ . Its variance is bounded as

$$\text{Var}[\hat{\theta}] \geq \frac{1}{\mathcal{I}(p(x|\theta))}, \quad (12.18)$$

where the quantity $\mathcal{I}(p(x|\theta))$ is called the *Fisher information* and defined as

$$\mathcal{I}(p(x|\theta)) = \int dx \frac{1}{p(x|\theta)} \left(\frac{\partial p(x|\theta)}{\partial \theta} \right)^2 = \int dx p(x|\theta) \left(\frac{\partial \ln p(x|\theta)}{\partial \theta} \right)^2 = E \left[\left(\frac{\partial \ln p(x|\theta)}{\partial \theta} \right)^2 \right]. \quad (12.19)$$

⁷⁷ This is probably the estimator that most people would use without giving further thought to the matter: flip the coin N times and take the ratio of Heads to N .

To prove the Cramér-Rao bound, we re-write the definition of an unbiased estimator in Eq (12.2) in a form convenient for us,

$$0 = \int dx p(x|\theta) (\hat{\theta}(x) - \theta). \quad (12.20)$$

Differentiate both sides with respect to θ to obtain

$$0 = \frac{\partial}{\partial \theta} \int dx p(x|\theta) (\hat{\theta}(x) - \theta) = \int dx \left(\frac{\partial p(x|\theta)}{\partial \theta} (\hat{\theta}(x) - \theta) - p(x|\theta) \right) \quad (12.21)$$

Using that $\int dx p(x|\theta) = 1$ we re-write the above as

$$1 = \int dx p(x|\theta) \left[\frac{\partial \ln p(x|\theta)}{\partial \theta} (\hat{\theta}(x) - \theta) \right], \quad (12.22)$$

where we have expressed the derivative from Eq (12.21) in a somewhat unintuitive way, but by expanding it we see that it is indeed valid. The reason we choose to write (12.21) on the form in Eq (12.22) is that it is suitable for applying the Cauchy-Schwarz inequality. Recall that on square-integrable functions the Cauchy-Schwarz inequality is $(\int dx g(x)h(x))^2 \leq \int dx g(x)^2 \times \int dx h(x)^2$. We use it in the following way: set $g(x) = \sqrt{p(x|\theta)} \frac{\partial \ln p(x|\theta)}{\partial \theta}$ and $h(x) = \sqrt{p(x|\theta)} (\hat{\theta}(x) - \theta)$. This leads to

$$1 \leq \underbrace{\int dx p(x|\theta) \left(\frac{\partial \ln p(x|\theta)}{\partial \theta} \right)^2}_{=\mathcal{I}(p(x|\theta)) \text{ via Eq (12.19)}} \times \underbrace{\int dx p(x|\theta) (\hat{\theta}(x) - \theta)^2}_{=\text{Var}[\hat{\theta}]}, \quad (12.23)$$

which can be re-arranged into the Cramér-Rao bound in Eq (12.18). $\square$

It may seem as if the definition of Fisher information is just a convenient name given to the factor appearing in Eq (12.23), but there is more to it than that. The Fisher information, as appearing in Eq (12.23), can be interpreted as the expectation value of $\left(\frac{\partial \ln p(x|\theta)}{\partial \theta} \right)^2$; see the right-most step in Eq (12.19). Here, we see the slope of the log-likelihood function appearing, namely $\frac{\partial \ln p(x|\theta)}{\partial \theta}$. If the slope is large then it should intuitively be easier to estimate θ : it narrows the values of θ associated with a high probability for the observed data sample. This is regardless of whether the slope is positive or negative, and indeed this is eliminated by the square in our expression. In conclusion, the Fisher information is giving us the average magnitude of the slope of the log-likelihood function, which is commonly interpreted as the amount of information about θ that can be extracted from the probability distribution⁷⁸.

This informational interpretation of $\mathcal{I}$ is further served by two properties: the Fisher information is non-negative and additive. Non-negativity follows immediately from the right-most expression in Eq (12.19) but additivity is less immediate. To prove additivity, consider that we take N independent samples, associated with the distribution $p(x|\theta) = \prod_{i=1}^N p(x_i|\theta)$. Then,

$$\begin{aligned} \mathcal{I}(p(x|\theta)) &\stackrel{(12.19)}{=} \int dx p(x|\theta) \left(\frac{\partial \ln p(x|\theta)}{\partial \theta} \right)^2 = \int dx \left(\prod_i p(x_i|\theta) \right) \left(\frac{\partial \ln \prod_j p(x_j|\theta)}{\partial \theta} \right)^2 \\ &= \int dx \left(\prod_i p(x_i|\theta) \right) \left(\sum_j \frac{\partial \ln p(x_j|\theta)}{\partial \theta} \right) \left(\sum_k \frac{\partial \ln p(x_k|\theta)}{\partial \theta} \right) \\ &= \sum_{j \neq k} \int dx \left(\prod_{i \neq j, i \neq k} p(x_i|\theta) \right) \frac{\partial p(x_j|\theta)}{\partial \theta} \frac{\partial p(x_k|\theta)}{\partial \theta} + \sum_{j=k} \int dx \left(\prod_{i \neq j} p(x_i|\theta) \right) \frac{1}{p(x_j|\theta)} \left(\frac{\partial p(x_j|\theta)}{\partial \theta} \right)^2. \end{aligned} \quad (12.24)$$

In the last line we have applied the chain rule of the derivative and for simplicity grouped terms according to $j \neq k$ and $j = k$. Also, note that $dx = dx_1 \dots dx_N$. In the first term on the last line, the integral over all x_i except $i = j$ and $i = k$ is simply $\int dx_i p(x_i|\theta) = 1$. Using this, the first term vanishes because

$$\int dx_j \frac{\partial p(x_j|\theta)}{\partial \theta} \int dx_k \frac{\partial p(x_k|\theta)}{\partial \theta} = \left(\frac{\partial}{\partial \theta} \underbrace{\int dx_j p(x_j|\theta)}_{=1} \right) \left(\frac{\partial}{\partial \theta} \underbrace{\int dx_k p(x_k|\theta)}_{=1} \right) = 0. \quad (12.25)$$

⁷⁸ In addition, one can also show that the Fisher information determines the lowest-order contribution to a suitable distance measure between $p(x|\theta)$ and $p(x|\theta + \delta\theta)$, for some small perturbation $\delta\theta$.

We are left only with the second term on the last line of Eq (12.24). For the same reason as above, when $i \neq j$ the integral over x_i is unity. It leaves us with

$$\mathcal{I}(p(x|\theta)) = \sum_j \int dx_j \frac{1}{p(x_j|\theta)} \left(\frac{\partial p(x_j|\theta)}{\partial \theta} \right)^2 \stackrel{(12.19)}{=} \sum_j \mathcal{I}(p(x_j|\theta)), \quad (12.26)$$

which is the sum of the Fisher information (12.19) over all the N random variables. Hence, the Fisher information is additive. For simplicity, consider that all the independent rounds are also identically distributed, namely $p(x_j|\theta) = p(x_1|\theta)$. Then, the Cramér-Rao bound implies that the variance of an unbiased estimator is limited by the scaling

$$\text{Var}[\hat{\theta}] \geq \frac{1}{N \times \mathcal{I}(p(x_1|\theta))}. \quad (12.27)$$

This scaling which is proportional to the inverse of N is known as the *standard quantum limit* or sometimes the *shot-noise limit*.

12.3. Quantum metrology

We are ready to treat parameter estimation in a quantum framework. To that end, consider that we prepare a multipartite state ρ and use a CPTP map Λ_θ to encode the parameter θ . Then, we perform a POVM $\{E_x\}_x$. The outcome is therefore sampled from the probability distribution

$$p(x|\theta) = \text{tr}(\Lambda_\theta[\rho]E_x). \quad (12.28)$$

Once we have our quantum probability distribution, we use our preferred estimator to infer θ just as before. Hence, the goal of quantum metrology is to optimise the quantum resources used in the sampling part when performing parameter estimation.

To emphasise the quantum nature of $p(x|\theta)$, we write the Fisher information as $\mathcal{I}(p(x|\theta)) = \mathcal{I}(\rho_\theta, E)$. Here, we label by $\rho_\theta = \Lambda_\theta[\rho]$ the parameter-encoded state. Since the Fisher information limits our precision, we want to find the POVM that maximises it, namely

$$\mathcal{I}^Q(\rho_\theta) \equiv \max_E \mathcal{I}(\rho_\theta, E) \stackrel{(12.19)}{=} \stackrel{(12.28)}{\max_E} \int dx \frac{\text{tr}(\dot{\rho}_\theta E_x)^2}{\text{tr}(\rho_\theta E_x)}. \quad (12.29)$$

where $\dot{\rho}_\theta = \frac{\partial \rho_\theta}{\partial \theta}$. This quantity, $\mathcal{I}^Q$, is called the *quantum Fisher information*. It represents the largest Fisher information achievable from a given quantum state and a given encoding rule for the parameter. This brings us to a quantum version of the Cramér-Rao bound.

R28: Quantum Cramér-Rao bound

Let θ be encoded in the state ρ via a CPTP map and let $\hat{\theta}(x)$ be an unbiased estimator for θ . The variance of the estimator is bounded as

$$\text{Var}[\hat{\theta}] \geq \frac{1}{\mathcal{I}^Q(\rho_\theta)}, \quad (12.30)$$

where the quantum Fisher information can be expressed as

$$\mathcal{I}^Q(\rho_\theta) = \sum_j \frac{\dot{\lambda}_j^2}{\lambda_j} + 2 \sum_{j,k} \frac{(\lambda_j - \lambda_k)^2}{\lambda_j + \lambda_k} |\langle \dot{\psi}_j | \psi_k \rangle|^2. \quad (12.31)$$

The bound in Eq (12.30) can always be saturated.

The maximisation appearing in Eq (12.29) for the quantum Fisher information might look like a formidable task but we can prove that it admits the closed form in Eq (12.31). To that end, we first need to introduce a handy concept: the *symmetric logarithmic derivative*, L_θ , is defined as the self-adjoint operator that satisfies

$$L_\theta \rho_\theta + \rho_\theta L_\theta = 2\dot{\rho}_\theta. \quad (12.32)$$

The reason that we make this definition is because we now can write

$$\mathrm{tr}(\dot{\rho}_\theta E_x) \stackrel{(12.32)}{=} \frac{1}{2} \mathrm{tr}(L_\theta \rho_\theta E_x) + \frac{1}{2} \mathrm{tr}(\rho_\theta L_\theta E_x) = \mathrm{Re}(\mathrm{tr}(L_\theta \rho_\theta E_x)), \quad (12.33)$$

where in the last step we used that $\mathrm{tr}(\rho_\theta L_\theta E_x) = \mathrm{tr}((\rho_\theta L_\theta E_x)^\dagger)^* = \mathrm{tr}(E_x L_\theta \rho_\theta)^* = \mathrm{tr}(L_\theta \rho_\theta E_x)^*$. The Fisher information therefore admits the following bound,

$$\begin{aligned} \mathcal{I}(\rho_\theta, E) &\stackrel{(12.29)}{=} \int dx \frac{\mathrm{Re}(\mathrm{tr}(L_\theta \rho_\theta E_x))^2}{\mathrm{tr}(\rho_\theta E_x)} \leq \int dx \left| \frac{\mathrm{tr}(L_\theta \rho_\theta E_x)}{\sqrt{\mathrm{tr}(\rho_\theta E_x)}} \right|^2 = \int dx \left| \mathrm{tr} \left(\underbrace{\frac{\sqrt{\rho_\theta} \sqrt{E_x}}{\sqrt{\mathrm{tr}(\rho_\theta E_x)}}}_{=A} \underbrace{\sqrt{E_x} L_\theta \sqrt{\rho_\theta}}_{=B} \right) \right|^2 \\ &\stackrel{\text{CS-ineq}}{\leq} \int dx \underbrace{\mathrm{tr}(A^\dagger A)}_{=1} \times \underbrace{\mathrm{tr}(B^\dagger B)}_{=\mathrm{tr}(L_\theta \rho_\theta L_\theta E_x)} = \mathrm{tr} \left(L_\theta \rho_\theta L_\theta \underbrace{\int dx E_x}_{=1} \right) = \mathrm{tr}(L_\theta^2 \rho_\theta). \end{aligned} \quad (12.34)$$

In the last line, we have used the Cauchy-Schwarz inequality for operators and then the completeness relation for POVMs. The bound on the right-hand-side is independent of the POVM and therefore applies to the quantum Fisher information. It is saturated by choosing $\{E_x\}_x$ as the eigenbasis of L_θ since this saturates the Cauchy-Schwarz inequality.

However, the bound in Eq (12.34) still depends on the symmetric logarithmic derivative. We can substitute it with a closed expression. To this end, motivated by Eq (12.32), let us define $\Omega(A) = \frac{1}{2} A \rho_\theta + \frac{1}{2} \rho_\theta A$. In Exercise C.4.2 it is shown that the inverse of Ω is

$$\Omega^{-1}(A) = \sum_{j,k} \frac{2}{\lambda_j + \lambda_k} A_{jk} |\psi_j\rangle \langle \psi_k|, \quad (12.35)$$

where $\rho_\theta = \sum_l \lambda_l |\psi_l\rangle \langle \psi_l|$ is the spectral decomposition of ρ_θ and $A_{jk} = \langle \psi_j | A | \psi_k \rangle$. By the definition of the symmetric logarithmic derivative in Eq (12.32) we have $\Omega(L_\theta) = \dot{\rho}_\theta$. Using Eq (12.35), we invert this to $L_\theta = \Omega^{-1}(\dot{\rho}_\theta)$. Inserting this in Eq (12.34) we obtain $\mathcal{I}^Q(\rho_\theta) = \mathrm{tr}(\Omega^{-1}(\dot{\rho}_\theta)^2 \rho_\theta)$. In Exercise C.4.2 we show that simplifying this expression gives the final result in Eq (12.31). $\square$

The quantum Fisher information has properties that are analogous to the ordinary Fisher information. Firstly, it is by definition non-negative. Secondly, it is additive in the sense that

$$\mathcal{I}^Q(\rho_\theta \otimes \sigma_\theta) = \mathcal{I}^Q(\rho_\theta) + \mathcal{I}^Q(\sigma_\theta). \quad (12.36)$$

Also, the quantum Fisher information is convex,

$$\mathcal{I}^Q(\lambda \rho_\theta + (1-\lambda) \sigma_\theta) \leq \lambda \mathcal{I}^Q(\rho_\theta) + (1-\lambda) \mathcal{I}^Q(\sigma_\theta), \quad (12.37)$$

for any θ -independent parameter $\lambda \in [0, 1]$. These properties are proven in Exercise C.4.3.

When Λ_θ is unitary, convexity implies that the maximal value of the quantum Fisher information taken over all density matrices is achieved for a pure state, $\rho = |\psi\rangle \langle \psi|$. The unitary encoding maps this to another pure state, $\rho_\theta = |\psi_\theta\rangle \langle \psi_\theta|$. It is therefore interesting to consider the quantum Fisher information for pure states. As shown in Exercise C.4.3, pure states simplify the expression in Eq (12.31) to

$$\mathcal{I}^Q(\psi_\theta) = 4 \left(\langle \dot{\psi}_\theta | \dot{\psi}_\theta \rangle - |\langle \dot{\psi}_\theta | \psi_\theta \rangle|^2 \right). \quad (12.38)$$

Unitary channels can be written as $U = e^{-i\theta H}$, where H is a Hermitian operator that can be thought of as the system's Hamiltonian. Our estimation task now is to learn the angle θ . Using the Schrödinger equation, namely $|\dot{\psi}\rangle = -iH|\psi\rangle$, the quantum Fisher information becomes

$$\mathcal{I}^Q(\psi_\theta) \stackrel{(12.38)}{=} 4 \left(\langle \psi | H^2 | \psi \rangle - \langle \psi | H | \psi \rangle^2 \right) = 4 \mathrm{Var}[H]. \quad (12.39)$$

Inserting this in the quantum version of the Cramér-Rao bound in Eq (12.30), we obtain

$$\mathrm{Var}[\hat{\theta}] \mathrm{Var}[H] \geq \frac{1}{4}. \quad (12.40)$$

This is an uncertainty relation between the estimator and the generator of the transformation that encodes the parameter. It is reminiscent of Robertson's uncertainty relation from Chapter 3.1.

12.4. The Heisenberg limit

Entanglement is the key quantum resource that will allow us to enhance parameter estimation beyond the standard quantum limit encountered in Eq (12.27). In other words, entanglement lets us achieve a better scaling in the variance than $\sim \frac{1}{N}$. To see how entanglement helps, consider that we prepare the GHZ state that we first encountered in Chapter 4.6. For N -partite systems it is defined as $|GHZ\rangle = \frac{1}{\sqrt{2}}(|0\rangle^{\otimes N} + |1\rangle^{\otimes N})$. We independently send each qubit through a unitary channel of the form $U_\theta = e^{i\frac{\theta}{2}}e^{-i\frac{\theta}{2}\sigma_z} = |0\rangle\langle 0| + e^{i\theta}|1\rangle\langle 1|$. Therefore, the state after the qubits are sent through the channel becomes

$$|\psi_\theta\rangle = (U_\theta \otimes \dots \otimes U_\theta)|GHZ\rangle = \frac{1}{\sqrt{2}}(|0\rangle^{\otimes N} + e^{iN\theta}|1\rangle^{\otimes N}). \quad (12.41)$$

Consider now that we measure all the N qubits separately using the observable $\sigma_X^{\otimes N}$. This effectively corresponds to measuring σ_X on each qubit, obtaining the outcome $a_i \in \{+1, -1\}$ from the measurement on the i 'th qubit and finally outputting the parity $a = \prod_i a_i$. The probability distribution over positive and negative parity becomes

$$p_\pm = \frac{1}{2} (1 \pm \langle \psi_\theta | \sigma_X^{\otimes N} | \psi_\theta \rangle) = \frac{1}{2} \left(1 \pm \frac{e^{iN\theta} + e^{-iN\theta}}{2} \right) = \frac{1 \pm \cos(N\theta)}{2}. \quad (12.42)$$

We compute the Fisher information of this distribution from Eq (12.19). Using that $\left| \frac{dp_\pm}{d\theta} \right| = \left| \frac{dp_-}{d\theta} \right|$, we obtain

$$\mathcal{I}(p_\pm) = \frac{1}{p_+} \left(\frac{dp_+}{d\theta} \right)^2 + \frac{1}{p_-} \left(\frac{dp_-}{d\theta} \right)^2 = \left(\frac{dp_+}{d\theta} \right)^2 \left(\frac{1}{p_+} + \frac{1}{p_-} \right) = \frac{4}{\sin^2(N\theta)} \times \frac{1}{4} N^2 \sin^2(N\theta) = N^2. \quad (12.43)$$

Furthermore, we would like to check that no larger Fisher information is possible. To that end, we must compute the quantum Fisher information. From Eq (12.34) we have that $\mathcal{I}^Q = \text{tr}(L_\theta^2 \rho_\theta)$, where $\rho_\theta = |\psi_\theta\rangle\langle\psi_\theta|$. Since ρ_θ is pure, we can determine the symmetric logarithmic derivative directly from Eq (12.32) as

$$L_\theta \rho_\theta + \rho_\theta L_\theta = 2 \frac{d\rho_\theta}{d\theta} \stackrel{\text{pure state}}{=} 2 \frac{d\rho_\theta^2}{d\theta} = 2(\dot{\rho}_\theta \rho_\theta + \rho_\theta \dot{\rho}_\theta) \Rightarrow L_\theta = 2\dot{\rho}_\theta. \quad (12.44)$$

Using the encoded state (12.41), we calculate that $\dot{\rho}_\theta^2 = \frac{N^2}{4} ((|0\rangle\langle 0|)^{\otimes N} + (|1\rangle\langle 1|)^{\otimes N})$. Hence, the quantum Fisher information is

$$\mathcal{I}^Q \stackrel{(12.34)}{=} \text{tr}(L_\theta^2 \rho_\theta) \stackrel{(12.44)}{=} 4 \text{tr}(\dot{\rho}_\theta^2 \rho_\theta) = N^2, \quad (12.45)$$

which is identical to Eq (12.43). We conclude that our measurement of the parity, $\sigma_X^{\otimes N}$, is optimal for parameter estimation. Thanks to our use of entanglement, we have achieved a quadratic scaling in N for the Fisher information, corresponding to the variance scaling as $\frac{1}{N^2}$. This is called the *Heisenberg limit*. It outperforms the $\frac{1}{N}$ scaling for the variance that we encountered for classical systems through the standard quantum limit.

Chapter summary.

- Estimation theory is the study of how to infer an unknown parameter from observed data. The Cramér-Rao bound imposes a limit on the best achievable precision for the estimation in terms of the Fisher information.
- For classical systems, the Fisher information scales linearly in the number of independent and identically distributed data samples.
- The use of multipartite entanglement makes possible a quadratic scaling for the Fisher information in the number of independent and identically distributed samples. This constitutes a quantum advantage in parameter estimation.

Further reading. For an extensive introduction, see for example the review paper [Quantum metrology from a quantum information science perspective](#) by Tóth and Apellaniz and Chapter 24 in [Modern Quantum Theory](#), by Bertlmann & Friis.